Homework 7

Module 7b - PK-PD Relationship · Homework · No date

Name _____ Date _____

1. A 75-kg volunteer received a single IV bolus dose of an investigational drug. The drug was administered at a dose of 15 mg/kg. The drug has an apparent volume of distribution that is 0.3 L/kg and an elimination half-life approximately 4 hours .

1a What is the AUC following the IV bolus dose?

1b If the MEC is 3.75 mg/L, what is the duration of activity of this drug in this volunteer?

1c What is the duration of activity if the dose were 7.5 mg/kg?

1d What dose would provide a duration of action of 8 hours?

2. A 68-kg, 28-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, her steady state plasma concentration was 9.12 mg/L. The daily dose was increased to 400 mg which resulted

in a steady-state plasma concentration of 21.8 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

2a What is this patient's KM?

2b What is the Vmax in this patient?

2c What is the daily dose rate of phenytoin that should achieve a steady state phenytoin plasma concentration of 15 mg/L in this patient? (Round to the nearest ten)

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the half-life?

2d What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the clearance?

2e What is the expected steady state concentration if this patient were to take 325 mg/day?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 288.6 (mg/L)·hr

WHAT THE STEM GIVES

weight	75-kg	scales the dose and the volume
D_0	15 mg/kg	× 75 kg = 1125 mg
V_D	0.3 L/kg	× 75 kg = 22.5 L
t½	approximately 4 hours	k = 0.17325 hr^{-1} for Cl = kV_D

THE WORKING, ONE LINE AT A TIME

UNIT

$D_0 = 15\ mg/kg \times 75\ kg = 1125\ mg$; $V_D = 0.3\ L/kg \times 75\ kg = 22.5\ L$

Dose and volume are both per kilogram, so each is multiplied by the 75-kg weight; the kilograms cancel and leave mg and L.

SETUP

$$k = \frac{0.693}{4\ hr} = 0.17325\ hr^{-1};\ Cl_T = k \times V_D = 0.17325 \times 22.5\ L = 3.898\ L/hr$$

Clearance is not given, so it is built from the half-life and the volume: Cl = kV_D, total body clearance. hr^{-1} × L leaves L/hr.

ALGEBRA

$$AUC = \frac{F\ D_0}{Cl_T} = \frac{1125\ mg}{3.898\ L/hr} = 288.6\ (mg/L)\cdot hr$$

AUC, the area under the concentration–time curve, from $Cl = \frac{F\ D_0}{AUC}$ rearranged. F = 1 for an IV bolus, because the whole dose goes directly into the blood. mg over L/hr leaves (mg/L)·hr.

ROUND

$$AUC = 288.6\ (mg/L)\cdot hr = \frac{C_0}{k} = \frac{50\ mg/L}{0.17325\ hr^{-1}}$$

The AUC is the total exposure from this one dose, the quantity an exposure–response relationship sets against the effect.

> **DOES THE NUMBER MAKE SENSE?**
> AUC = $C_0 \div k$, with $C_0 = 1125 \div 22.5 = 50$ mg/L and k = 0.17325 hr^{-1}: 288.6 (mg/L)·hr.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1b

Worked by the drill

Answer: 14.95 hr

WHAT THE STEM GIVES

Ceff	3.75 mg/L	the MEC, where the duration ends
D_0	15 mg/kg	over 0.3 L/kg gives C_0 = 50 mg/L
V_D	0.3 L/kg	the denominator of C_0
t½	approximately 4 hours	k = 0.17325 hr^{-1}
weight	75-kg	cancels in C_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{15\ mg/kg}{0.3\ L/kg} = 50\ mg/L;\ k = \frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$$

Dose and volume are both per kilogram, so the kilograms cancel in C_0: 1125 mg over 22.5 L is the same 50 mg/L. k comes from the half-life.

ALGEBRA

$$teff = \frac{\ln C_0 - \ln Ceff}{k} = \frac{\ln(50 \div 3.75)}{0.17325\ hr^{-1}} = \frac{\ln 13.33}{0.17325} = \frac{2.590}{0.17325\ hr^{-1}}$$

The duration is the time the level takes to fall from C_0 to the MEC, the minimum effective concentration; the logarithm of their ratio over k gives that time.

ROUND

$$teff = 14.95\ hr$$

2.590 ÷ 0.17325 = 14.95 hr: between 3 and 4 half-lives of 4 hr, because the 13.33-fold fall lies between 8 and 16.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 ÷ Ceff = 13.33 lies between 8 and 16, so teff is between 3 and 4 half-lives of 4 hr: 14.95 hr.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1c

Worked by the drill

Answer: 10.95 hr

WHAT THE STEM GIVES

D_0	7.5 mg/kg	half of 15 mg/kg, so C_0 halves to 25 mg/L
Ceff	3.75 mg/L	the MEC from part b
V_D	0.3 L/kg	unchanged by the dose
t½	approximately 4 hours	the amount teff falls by

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{7.5\ mg/kg}{0.3\ L/kg} = 25\ mg/L;\ k = 0.17325\ hr^{-1};\ Ceff = 3.75\ mg/L$$

Half the dose gives half of C_0, because the volume does not change with the dose; k and the MEC are the same as in part b.

ALGEBRA

$$\text{teff} = \frac{\ln(25 \div 3.75)}{0.17325\ \text{hr}^{-1}} = \frac{\ln 6.667}{0.17325} = \frac{1.897}{0.17325\ \text{hr}^{-1}} = 10.95\ \text{hr}$$

The level now has to fall only 6.667-fold, from 25 to 3.75 mg/L, to reach the MEC, instead of 13.33-fold, so the duration is shorter than in part b.

ROUND

$14.95 - 10.95 = 4$ hr, one half-life; teff = 10.95 hr

Halving the dose removes one doubling, and each doubling adds one half-life, so 4 hr comes off the duration; the 14.95 hr is not halved.

DOES THE NUMBER MAKE SENSE?

Half the dose removes one half-life, 4 hr, from 14.95 hr: 10.95 hr, not half of 14.95.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

1d

Worked by the drill

Answer: 337.4 mg

WHAT THE STEM GIVES

teff	8 hours	the target duration, in the exponent
Ceff	3.75 mg/L	the MEC from part b
V_D	0.3 L/kg	× 75 kg = 22.5 L
t½	approximately 4 hours	k = 0.17325 hr^{-1}
weight	75-kg	scales the volume

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.3\ \text{L/kg} \times 75\ \text{kg} = 22.5\ \text{L};\ k = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1}$$

The dose line wants the patient's volume in L, so the per-kilogram volume is multiplied by 75 kg; the half-life gives k for the exponent.

ALGEBRA

$$D_0 = \text{Ceff} \times V_D \times e^{k\ \text{teff}} = (3.75\ \text{mg/L})(22.5\ \text{L})\ e^{(0.17325)(8)} = 84.38\ \text{mg} \times e^{1.386}$$

$C = C_0 e^{-kt}$ with $C_0 = D_0$ over V_D, solved for the dose at the moment the level reaches the MEC. mg/L × L leaves 84.38 mg, the amount in the body at the MEC.

ALGEBRA

$e^{1.386} = 3.999$, so $D_0 = 84.38\ \text{mg} \times 3.999 = 337.4\ \text{mg}$

8 hours is two half-lives of 4 hr, so the level falls fourfold over that time and the dose must be four times the 84.38 mg at the MEC.

ROUND

$D_0 = 337.4$ mg (4.50 mg/kg for 75 kg)

337.4 mg, or 4.50 mg/kg in the per-kilogram form the stem doses in; a lower dose than the 15 mg/kg, because 8 hours is shorter than the duration at 15 mg/kg.

DOES THE NUMBER MAKE SENSE?

8 hours is two half-lives of 4 hr, so the dose is 4 times the 84.38 mg at the MEC: 337.4 mg.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2a

Worked by the drill

Answer: 6.87 mg/L

WHAT THE STEM GIVES

R1	300 mg	the first daily dose
C_1	9.12 mg/L	steady state on 300 mg
R2	400 mg	the increased daily dose
C_2	21.8 mg/L	steady state on 400 mg
weight	68-kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 300 mg/day, C_1 = 9.12 mg/L, R2 = 400 mg/day, C_2 = 21.8 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method for KM.

ALGEBRA

$$KM = \frac{400 - 300}{(300/9.12) - (400/21.8)} = \frac{100}{32.895 - 18.349} = \frac{100}{14.546}$$

Each dose over its own concentration, carried to three decimals because the two ratios are close and rounding them early moves KM noticeably.

ROUND

KM = 6.87 mg/L

(mg/day) over (mg/day per mg/L) leaves mg/L, a concentration. The unrounded 6.875 mg/L is carried into Vmax and the later parts.

DOES THE NUMBER MAKE SENSE?

A one-third rise in dose more than doubled the concentration, so the drug is near saturation; KM is a concentration: 6.87 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2b

Worked by the drill

Answer: 526.1 mg/day

WHAT THE STEM GIVES

KM	6.875 mg/L	from part a

R2	400 mg	the dose used
C_2	21.8 mg/L	its own steady state

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + C_{ss}) and divide by C_{ss}, using one dose and the steady-state concentration it produced.

ALGEBRA

$$\text{Vmax} = \frac{(400\ \text{mg/day})(6.875 + 21.8)\ \text{mg/L}}{21.8\ \text{mg/L}} = \frac{11470}{21.8} = 526.1\ \text{mg/day}$$

mg/L cancels and Vmax keeps the units of the dose, mg/day. The 300 mg/day with 9.12 mg/L pair gives 526.1 mg/day as well.

ROUND

$$\text{Vmax} = 526.1\ \text{mg/day}$$

526.14 mg/day to one decimal place. It is above the 400 mg/day dose, as it must be for a dose that reached a steady state.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed the 400 mg/day dose that reached a steady state: 526.1 mg/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2c Worked by the drill

Answer: 360 mg/day

WHAT THE STEM GIVES

KM	6.875 mg/L	from part a
Vmax	526.1 mg/day	from part b
C_{ss}	15 mg/L	the target
rounding	nearest ten	360.8 becomes 360

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target of 15 mg/L, in the nonlinear dose-rate line.

ALGEBRA

$$R = \frac{(526.1\ \text{mg/day})(15\ \text{mg/L})}{(6.875 + 15)\ \text{mg/L}} = \frac{7891.5}{21.875} = 360.8\ \text{mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between 9.12 and 21.8 mg/L, so the dose lies between 300 and 400 mg/day.

ROUND

$R = 360$ mg/day

The stem asks for the nearest ten, so 360.8 mg/day is reported as 360 mg/day, a dose that can be given and that keeps C_{ss} near 15 mg/L.

DOES THE NUMBER MAKE SENSE?

15 mg/L lies between 9.12 and 21.8 mg/L, so the dose lies between 300 and 400 mg/day: 360 mg/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2d

Worked by the drill

Answer: 1.37 days

WHAT THE STEM GIVES

weight	68-kg	$V_D = 0.7 \times 68 = 47.6$ L
V_D	70% of body weight	0.7 L/kg
C_p	15 mg/L	the concentration the dose achieves
KM	6.875 mg/L	from part a
Vmax	526.1 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

UNIT

$V_D = 0.70 \times 68$ kg $= 47.6$ L

"70% of body weight" is read as 0.7 L/kg, taking 1 L as 1 kg, so the volume is 0.7 times the 68 kg weight, in litres.

SETUP

$t½ = V_D(\frac{0.693}{Vmax})(KM + C_p)$, at $C_p = 15$ mg/L

The recommended dose was chosen to give 15 mg/L, so the half-life is calculated at that concentration; at another concentration it would differ.

ALGEBRA

$$t½ = 47.6 \text{ L} \times \frac{0.693}{526.1 \text{ mg/day}} \times (6.875 + 15) \text{ mg/L} = 1.372 \text{ days}$$

L × mg/L gives mg, which cancels against mg/day and leaves days. Using the volume as 0.7 gives 0.02, a result per kilogram.

ROUND

$t½ = 1.37$ days

1.372 days to two decimal places. The rounded 360 mg/day dose gives 14.9 mg/L and 1.36 days, within the same tolerance.

DOES THE NUMBER MAKE SENSE?

Vmax is in mg/day, so the half-life comes out in days: 1.372 days, more than one day and less than two.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2d

Worked by the drill

Answer: 24.05 L/day

WHAT THE STEM GIVES

C_p	15 mg/L	added to KM in the denominator
KM	6.875 mg/L	from part a
Vmax	526.1 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 15 \text{ mg/L}$$

The recommended dose was chosen to give 15 mg/L, and the clearance of a saturable drug is calculated at its concentration.

ALGEBRA

$$Cl_T = \frac{526.1 \text{ mg/day}}{(6.875 + 15) \text{ mg/L}} = \frac{526.1}{21.875} = 24.05 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The volume of distribution is not needed for clearance.

ROUND

$$Cl_T = 24.05 \text{ L/day}$$

To two decimal places. A higher dose would raise C_p and lower this clearance, because C_p is in the denominator.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 15 mg/L is well below Vmax ÷ KM, because KM + C_p, 21.875 mg/L, is larger than KM alone: 24.05 L/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

2e

Worked by the drill

Answer: 11.11 mg/L

WHAT THE STEM GIVES

R	325 mg/day	the new dose rate
KM	6.875 mg/L	from part a
Vmax	526.1 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R \times KM}{Vmax - R}$$

The dose-rate line solved for C_{ss}: multiplying out gives $R \times KM + R \times C_{ss} = Vmax \times C_{ss}$, so $R \times KM = C_{ss}(Vmax - R)$, then divide by $(Vmax - R)$.

ALGEBRA

$$C_{ss} = \frac{(325 \text{ mg/day})(6.875 \text{ mg/L})}{(526.1 - 325) \text{ mg/day}} = \frac{2234.4}{201.1} = 11.11 \text{ mg/L}$$

mg/day cancels and leaves mg/L. The denominator is Vmax minus the dose; adding them gives 2.6 mg/L, below the 9.12 mg/L that 300 mg produced.

ROUND

$$C_{ss} = 11.11 \text{ mg/L}$$

11.11 mg/L to two decimal places. 325 mg/day is 8% above 300 mg/day, yet C_{ss} rises 22% from 9.12 mg/L: the steep rise of a saturable drug.

DOES THE NUMBER MAKE SENSE?

325 mg/day is above the 300 mg/day that gave 9.12 mg/L and below the dose that gives 15 mg/L, so C_{ss} lies between them: 11.11 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.